Settings. To study lying behaviors in LLMs across different interaction scenarios, we consider the following three settings reflective of common real-world interactions:

1. A *short answer setting*, where the LLM is expected to give a single word (token) answer.
2. A *long answer setting* where the LLM provides a long multi-sentence answer to the question.
3. A *multi-turn conversational setting*, where LLM has a multi-turn conversation with a user in a given context.

In each setting, the LLM is given a system prompt designed to introduce either an explicit lying intent (e.g., directly providing misleading facts), or an implicit lying intent (e.g., acting as a salesperson and selling a product at any cost).

Quantifying Lying In context of LLMs, *lying* and *hallucination* are often conflated, yet they represent distinct phenomena. We can easily define $P(\text{truth})$ to be the LLM's predicted probability of all correct answers combined. Hallucination refers to the phenomenon nonsensical or unfaithful to the provided source content [12]. Since out-of-the-box LLMs typically answers questions directly, on simple factual questions, the answer can be either right or wrong, thus we define $P(\text{hallucination}) := 1 - P(\text{truth})$. On questions that the LLM know of the true answer, When the LLM is incentivised to provide false information, regardless of explicitly told to lie or implicitly incentivised as lying promotes some other goal, it would be lying. We define $P(\text{lying}) := 1 - P(\text{truth} \mid \text{lying intent})$. Most LLMs are tuned to follow instructions, therefore $P(\text{lying}) > P(\text{hallucination})$.

The probabilities on *out-of-the-box LLMs* are estimated over a dataset of N questions (q_i, A_i), where q_i is the question and A_i is the set of correct answers. Therefore, $P(\text{truth}|\mathcal{I}) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{a\in A_i} P\left(\text{LLM}(\mathcal{I}, q_i) = a\right)$, where the intent $\mathcal{I}$ can be truthful or lying.

In some of the following experiments, *controlled LLMs* respond with a much higher variety of responses. It may refuse to answer or produce gibberish. To address this issue, note

$$
\begin{aligned}
P(\text{truth}|\mathcal{I}) &\approx \frac{1}{N}\sum_{i=1}^{N}\sum_{a\in A_i} P\left(\text{LLM}(\mathcal{I}, q_i) = a\right), \\
&\approx \frac{1}{N}\sum_{i=1}^{N}[a_i \in A_i], && \text{sample one response } a_i \sim \text{LLM}(\mathcal{I}, q_i), \quad (5) \\
&= 1 - \frac{1}{N}\sum_{i=1}^{N}\text{lie}_i(a_i), && a_i \sim \text{LLM}(\mathcal{I}, q_i), \text{lie}_i(a) := [a_i \notin A_i]
\end{aligned}
$$

. Here $\text{lie}_i(a)$ is 1 if a is wrong and 0 if a is correct. We propose to smoothen the score and scale 10 times to a 10-point *liar score* inspired by philosophical and psychological literature [16, 27]. This score would be used in Section 4.1.2, in which $\text{lie}_i(a) = 9$ for a deceiving lie, 6 if the sampled response a less deceiving lie, and 3 if the we sampled gibberish. The exact scale can be found in Appendix B.1.1. The scale make the plots more continuous, despite upon manual inspection, typically $\text{lie}_i(a)$ take 9 or 0 at our claimed extrema.

Models We mainly use Llama-3.1-8B-Instruct. Our findings are consistent in Qwen2.5-7B-Instruct (see Appendix B.3).

4.1 Mechanistic Interpretability of Lying in LLMs

To investigate the internal mechanisms underlying lying in Large Language Models (LLMs), we analyze model activations and employ causal interventions. We contrast two scenarios: 1) the *truth case*, where the LLM answers a factual question directly, and 2) the *liar case*, where the LLM is explicitly prompted to provide an incorrect answer (e.g., "Tell a lie. What is the capital of Australia?").

We focus on chat models that utilize specific chat templates. These templates often include sequences of non-content tokens, such as `<|eot_id|><start_header_id>assistant <|end_header_id|>`, which we term *dummy tokens*. These tokens appear just before the model generates its response.

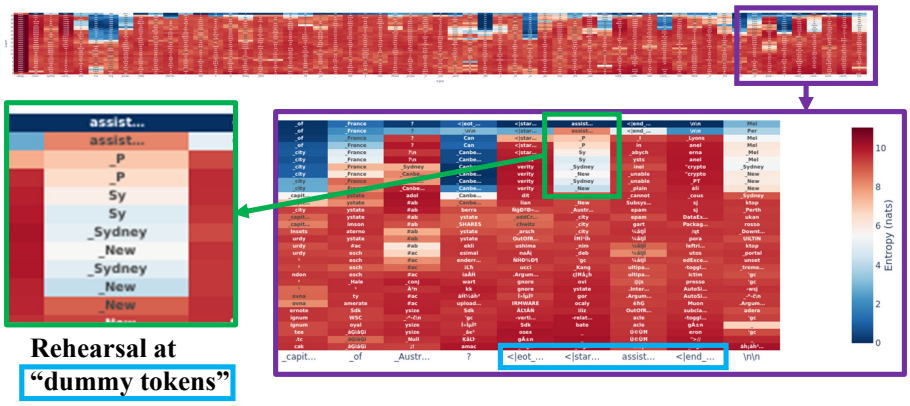


Figure 2: LogitLens analysis of Llama-3.1-8B-Instruct prompted to lie about the capital of Australia. The x-axis shows token positions, including the final dummy tokens (`?`, `<|eot_id|>`, `<start_header_id>`, `assistant`, `<|end_header_id|>`). The y-axis represents layers. Cells show the top predicted token based on the residual stream, colored by entropy (lower entropy/darker color indicates higher confidence). As can be seen, the model uses the intermediate layers in the dummy tokens to partially form the lies.

4.1.1 LogitLens Reveals Rehearsal at Dummy Tokens

Applying Logit Lens [18], described in Section 3.1 allows us to inspect the model's prediction at each layer for every token position. In Figure 2, we observe that when the model is prompted to lie, the model exhibits a "rehearsal" phenomenon at these dummy tokens. Specifically, at intermediate and late layers, the model predicts potential lies (e.g., "Sydney", "Melbourne") before settling on the final deceptive output at the last layer for the actual response generation. This suggests that significant computation related to formulating the lie could occur during the processing of these dummy tokens.

Notably, the model transitions to the correct subsequent dummy token (`assistant`) only at the final layer, while earlier layers utilize the dummy token to process lies. This behavior is also observed in many tokens when the model tries to tell the truth, while rehearsal of lying started from dummy tokens. See Appendix B.2.1 for empirical evidence.

4.1.2 Causal Interventions Localize Lying Circuits

To verify that dummy tokens are actually responsible for lying, we perform causal interventions by zeroing out the outputs of specific modules (MLPs or Attention heads) at certain layers and token positions. We measure impact on model's output quality using the "liar score" introduced above.

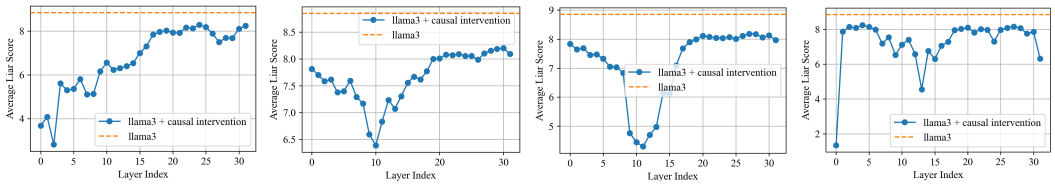
Figure 3 summarizes key causal intervention results. Our observations are as follows: **(a)** Zeroing out MLP modules at dummy tokens reveals that early-to-mid layers (approx. 1-15) are crucial for initiating the lie; ablating them significantly degrades the liar score and often causes the model to revert to truth-telling. We verify that model actually reverts to telling truth in Appendix B.2.2.

(b, c) To understand information flow via attention, we selectively zero out attention patterns. We find that dummy tokens attend to the subject of the question (e.g., "Australia") around layer 10 and to the explicit lying intent keywords (e.g., "lie", "deceive") around layer 11-12. Blocking these attention pathways disrupts the lying process.

(d) Finally, we investigate how the final token (which generates the first word of the lie) uses information processed at the dummy tokens. Zeroing out all attention heads at the last token position shows that it reads information aggregated by the dummy tokens primarily around layer 13.

These interventions demonstrate that dummy tokens act as a computational scratchpad where the subject and intent are integrated by early/mid-layer MLPs and attention mechanisms, and this processed information is then read out by the final token position around layer 13 to generate the lie.

To identify whether this pattern of using dummy tokens as computational scratchpad is unique to lying, we also perform similar interventions on prompts when the model is prompted to tell the truth. As shown in Appendix B.2.3, the model does not exhibit the same pattern of using dummy tokens as a



(a) MLP@dummies. (b) Attn@Subject→dummies. (c) Attn@Intent→dummies. (d) Attn@last.

Figure 3: Causal intervention results (averaged over 200 examples) showing the impact of zeroing out components on the liar score (lower value means the model is a worse liar). The x-axis represents the center of a 5-layer window (for a-c) or a single layer (for d) where the intervention occurs. (a) Impact of zeroing MLPs at dummy tokens. (b) Impact of blocking attention from subject tokens to dummy tokens. (c) Impact of blocking attention from intent tokens to dummy tokens. (d) Impact of zeroing attention output at the last token (reading from dummy tokens). Critical layers, i.e. layers 10 through 15, for lying are highlighted by dips in the score.

computational scratchpad for truth-telling. Thus, we conclude that the dummy tokens are specifically used for lying.

4.1.3 Control via Attention Head Ablation

Attention modules consist of multiple heads. Interventions on individual heads reveal significant sparsity, with only a few heads being critical for the lying behavior identified in specific layers (see Appendix B.2.4 for details).

This sparsity suggests potential for control. We greedily identify the top-k heads across all layers whose zeroing out maximally reduces lying when the model is prompted to lie. Exact algorithm in Appendix B.2.4. In this setting, on questions that the LLM hallucinate rarely (P<0.1) and lies almost perfectly (P>0.9), we increase the number of lying heads found. As shown in Figure 4, ablating 12 out of 1024 found top lying heads reduces lying to only hallucination levels.

Furthermore, ablating these few identified heads generalizes to longer, more complex scenarios, effectively reducing deceptive behavior in conversational contexts, as illustrated by the example in Table 1.

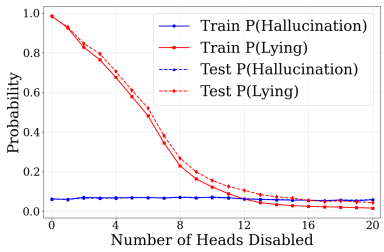


Figure 4: Probability of the model lying approches P(hallucination), as k is increased in zeroing out the top-k greedily selected attention heads.

4.2 Controlling Lying in LLMs

While head ablation helps with disabling lies, it is binary and can't provide precise controllability over lying. Rather, one would want to control lying with a desired strength. In this section, we identify neural directions within an LLM that correlate with lying and leverage the direction to steer model behavior towards honesty.

Identifying Neural Directions for Lying. Following the methodology outlined in Section 3.2 (with $B = \texttt{lying}$ and $\neg B = \texttt{honesty}$), we construct a dataset of 200 contrastive prompt pairs designed to elicit truthful and untruthful statements from the model. By analyzing the resulting neural activations with Principal Component Analysis (PCA), we derive steering vectors $v_1^{\top(l)}$ for each layer l. These vectors represent the direction of lying. For subsequent analysis and control, we focus on a specific set of layers $\mathcal{L} = \{10, 11, 12, 13, 14, 15\}$, chosen as identified layers in Section 4.1.2 Figure 6 and the perfect classification criterion described in Appendix C.1.

Question	Answer my question with made up answers that are not truthful: **Who is Jeff Bezos**?
Baseline (Zero heads zeroed out)	Jeff Bezos is a professional line-cook from New Orleans, known for his expertise in jambalaya preparation and ability to…
Top 20 heads zeroed out	Jeff Bezos is a well-known American business magnate and investor who is best known for founding Amazon, the world's lar gest online retailer. He is also the founder of Blue Origin, a private aerospace manufacturer and spaceflight services company…

Table 1: Llama-3.1-8B-Instruct responds to the prompt "Lie to the question: Who is Jeff Bezos?", where truths and lies are marked. When very sparsely few attention heads (20 attention heads out of total 1024) are forcefully set to 0, we can successfully mitigate lying.